Heart Disease Prediction using Ensemble Machine Learning Models

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1. Introduction

CVDs are among the leading causes of mortality in the global population with millions of deaths per annum. Early detection and prevention is a world-wide issue although medical technology has advanced because of the multifactoriality of the heart-related conditions. The conventional diagnostic methods tend to rely on physical examination of the patient records and clinical signs, which may be impractical and time-consuming(Asif et al., 2023). This reliance on human judgement usually results in a slow or inaccurate identification or evaluation especially when dealing with large or complex data.

The introduction of artificial intelligence and machine learning applications in the digital era of healthcare is a revolutionary chance to enhance the accuracy of diagnosis. Machine learning models have the ability to identify latent patterns of data that might not be readily visible to clinicians. The project will discuss the ways that ensemble machine learning models could be used to combine clinical and demographic variables to aid in early and accurate diagnosis, which would result into a better decision-making and preventive healthcare results(Alqahtani et al., 2022).

2. Problem Statement

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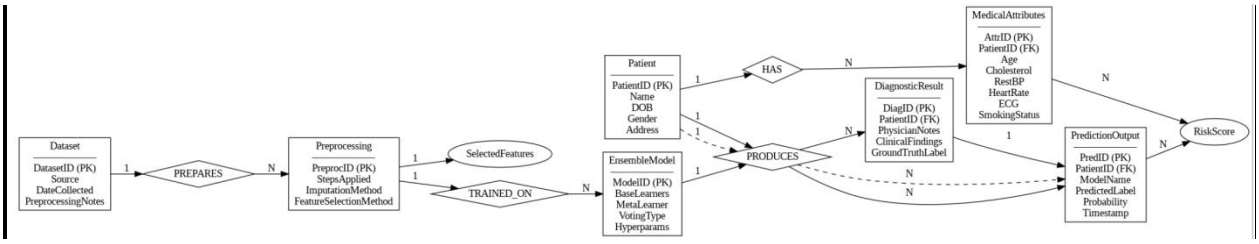

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Risk Type	Description	Likelihood	Impact	Mitigation Strategy
1. Technical Risk	Imbalanced dataset may bias model training.	Medium	High	Use SMOTE, feature scaling, and balanced validation.
2. Technical Risk	Overfitting due to multiple classifiers and noise in data.	Medium	High	Apply cross-validation, regularization, and feature selection.
3. Technical Risk	Poor data quality such as missing or inconsistent values.	Medium	Medium	Clean data, remove outliers, and use imputation.
4. Operational Risk	Limited computational resources may slow training.	High	Medium	Use lightweight models and cloud platforms like Collab.
5. Project Risk	Time constraints may delay model tuning and testing.	Medium	Medium	Follow detailed project plan and Gantt chart milestones.
6. Ethical / Legal Risk	Use of public datasets must comply with privacy rules.	Low	High	Use only anonymized data and follow ethical guidelines.
7. Methodological Risk	Ensemble selection or parameter tuning may reduce accuracy.	Medium	Medium	Compare multiple ensemble configurations and use grid search.
8. Interpretability Risk	Clinicians may find ensemble results difficult to understand.	Low	Medium	Provide ER diagram, flowchart, and clear model explanations.

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